

CALEDONIA COUNTY AP, VT, USA

WMO: 720492

Lat: 44.567N

Lon: 72.017W

Elev: 1188

StdP: 14.08

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00151

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.8	-5.1	-18.9	2.0	-7.8	-14.3	2.6	-1.7	20.9	24.7	19.2	20.7	3.6	230	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	84.0	68.7	81.4	66.5	78.9	65.8	71.3	79.7	69.5	77.2	68.1	75.6	7.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	109.4	75.2	67.1	104.1	73.8	65.4	98.0	71.8	35.9	80.0	34.4	77.5	33.2	75.8	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	16.0	13.0	-19.6	88.6	4.5	1.8	-22.8	89.8	-25.4	90.9	-27.9	91.9	-31.1	93.2		
			-19.3	74.6	4.7	1.5	-22.6	75.6	-25.4	76.5	-28.0	77.3	-31.4	78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	17.3	19.9	28.2	41.5	54.4	61.4	67.1	65.4	58.3	46.3	34.5	23.1
	DBStd	19.65	12.03	11.00	10.98	9.12	7.78	6.27	5.08	5.12	7.60	8.27	9.33	10.95
	HDD50	4358	1015	843	679	286	42	2	0	0	13	174	471	833
	HDD65	8187	1480	1263	1141	706	340	142	36	59	226	580	916	1298
	CDD50	1890	0	0	3	31	177	344	531	478	262	59	5	0
	CDD65	246	0	0	0	1	10	35	102	72	25	1	0	0
	CDH74	1972	0	0	1	22	131	317	805	498	194	4	0	0
	CDH80	352	0	0	0	4	20	61	165	70	32	0	0	0
Wind	WSAvg	4.7	4.9	5.4	6.1	6.3	5.1	4.2	3.4	3.2	3.4	4.6	4.9	4.7
Precipitation	PrecAvg	45.7	3.1	2.7	3.1	3.2	3.6	4.3	5.1	5.0	3.6	4.6	3.7	3.8
	PrecMax	53.6	5.9	4.8	4.9	6.7	6.7	7.7	8.0	9.9	7.8	8.3	5.3	7.4
	PrecMin	34.9	1.1	1.2	1.5	0.9	1.7	1.5	3.0	2.3	2.0	1.4	2.2	2.0
	PrecStd	5.0	1.0	0.9	0.9	1.5	1.3	1.7	1.2	1.9	1.3	2.1	0.9	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	51.7	66.4	77.1	83.0	85.9	88.2	84.6	84.5	72.4	62.6	48.1
		MCWB	47.0	45.1	56.4	57.9	63.6	68.5	71.0	70.2	70.6	60.6	53.0	45.9
	2%	DB	41.3	43.5	53.6	68.7	78.0	81.4	84.3	81.8	79.1	67.4	57.0	44.8
		MCWB	37.6	39.0	43.3	54.1	60.5	65.4	69.6	67.9	66.8	57.2	52.0	42.5
	5%	DB	36.9	39.0	46.4	63.0	73.3	77.5	81.5	79.3	75.7	63.3	53.4	40.9
		MCWB	34.8	35.1	39.4	51.0	59.0	63.8	67.0	67.0	64.9	56.1	48.3	38.7
	10%	DB	33.5	35.5	43.1	56.9	69.6	74.1	78.9	76.9	71.9	59.6	48.4	37.0
		MCWB	31.8	32.5	36.9	45.8	56.4	61.8	66.4	65.7	63.4	53.4	43.8	34.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	47.8	54.8	60.7	67.5	71.4	74.3	73.1	71.4	64.4	56.7	46.5
		MCDB	48.0	50.3	66.5	69.2	79.3	80.2	83.6	81.1	80.9	68.1	58.6	47.1
	2%	WB	38.8	39.7	45.8	57.0	64.4	69.1	72.0	70.4	69.0	60.2	52.8	42.9
		MCDB	40.5	42.7	50.8	67.4	73.9	78.1	80.4	78.1	76.4	64.3	56.4	44.8
	5%	WB	34.7	35.2	40.8	52.3	61.7	66.6	70.1	68.7	67.1	57.3	48.6	39.0
		MCDB	36.9	38.5	45.9	59.9	68.9	74.2	77.7	75.8	73.6	62.5	52.4	40.9
	10%	WB	31.5	32.4	37.5	47.6	59.0	64.1	68.3	67.3	64.7	54.2	44.1	35.2
		MCDB	33.0	34.7	42.4	55.7	66.3	70.7	75.8	73.9	70.2	58.8	47.7	36.7
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.7	16.9	18.3	20.6	19.5	20.0	20.5	20.4	16.4	14.1	13.2
		MCDBR	15.7	18.7	21.5	29.3	27.3	24.8	23.5	22.5	24.3	21.5	21.3	14.7
		MCWBR	15.1	15.1	13.9	16.5	13.5	12.6	11.0	11.3	12.6	13.7	15.4	14.3
	5% WB	MCDBR	15.0	18.0	18.5	25.4	20.4	19.8	19.1	18.8	21.3	18.7	17.9	13.9
		MCWBR	15.0	15.6	13.3	16.5	11.7	11.3	10.3	10.2	12.3	13.5	15.6	14.0
Clear-Sky Solar Irradiance	taub		0.257	0.276	0.311	0.345	0.363	0.411	0.420	0.393	0.356	0.321	0.301	0.268
	taud		2.405	2.325	2.350	2.396	2.406	2.262	2.247	2.313	2.415	2.524	2.508	2.442
	Ebn at Noon		279	291	293	291	288	272	268	272	275	272	260	262
	Edh at Noon		23	30	34	36	36	42	42	38	32	25	21	20
All-Sky Solar Radiation	RadAvg		449	724	1085	1349	1560	1705	1730	1564	1264	745	463	334
	RadStd		55	81	72	164	180	150	138	85	93	79	61	28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	N/A

Nomenclature: See separate page